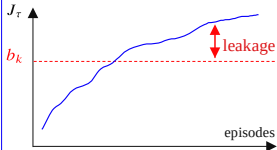


## Penalty view

$$\max_{\tau} J_{\tau} - \lambda^{\top} J_c$$

costs  $\rightarrow$  objective  
 $\tau \rightarrow \infty$

$$J_{\tau} = [J_{c1}, \dots, J_{cN}]^{\top}, \quad \lambda = [\lambda_1, \dots, \lambda_N]^{\top}$$



fixed- $\lambda$  penalty: some safety tasks reward  $b_k$

weakness: fixed-price leakage

## SafeOT

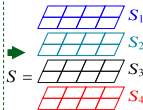
### 1. Constraints

|                |                 |
|----------------|-----------------|
| $C_1$<br>$S_1$ | $C_1 \cdot b_1$ |
| $C_2$<br>$S_2$ | $C_2 \cdot b_2$ |
| $C_3$<br>$S_3$ | $C_3 \cdot b_3$ |
| $H$<br>$S_4$   | $H: F\_H = 0$   |

### 2. One flow

$$\text{one flow}$$

$$S = \min_i S_i$$



$$F\star \rightarrow \lambda\star$$

### 3. Actor (entropic)

$$\lambda\star + \beta$$

$$\rightarrow \text{actor}$$

$$P =$$

|             |             |             |
|-------------|-------------|-------------|
| $\lambda_1$ | $\lambda_2$ | $\lambda_3$ |
| $b_1$       | $b_2$       | $b_3$       |
| $F\star$    | $\beta$     | $\lambda$   |

safety prices

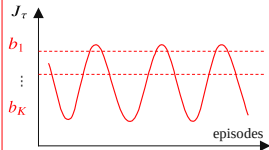
$$P^{\top} \leq S$$

## Constraint

$$\max J_{\tau} \quad \text{s.t.} \quad J_c \leq b$$

$$\tau \rightarrow 0$$

$$J_{\tau} = [J_{c1}, \dots, J_{cN}]^{\top}, \quad b = [b_1, \dots, b_N]^{\top}$$



dual updates: safety costs oscillate around  $b_k$

weakness: late / oscillation

$\tau = \infty$

fixed

Lagrangian

one price rule

SafeOT

LP  $\varepsilon \rightarrow 0$  / active

$\tau = 0$

penalty

constraints